

Learning Life Satisfaction

Predicting Subjective Well-Being with Linear and Non-Linear Methods

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Statistical Learning and Large Data 1 & 2



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Context

Life-satisfaction and subjective wellbeing are the object of study of a long-standing stream of literature.

While income has always been considered a reliable proxy of wellbeing, recent contributions highlighted that a broader set of complementary factors including health, social relationships, time allocation, and living conditions are likely to play a role.

Moreover, the literature increasingly questioned the linearity and stability of the income–happiness relationship: the Easterlin paradox (Easterlin, 1974) and the context-dependent evidence proposed by Alesina et al. (2004) suggest that the effect of income on wellbeing may vary across social and economic environments.

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Objective



The present study focuses on the prediction of individuals life satisfaction using both economic and non-economic covariates as explanatory variables.

The aim is purely predictive, with causal interpretations lying beyond the scope of the analysis, and the focus is twofold:

- Understanding how far linear models can be pushed in terms of shrinkage, variable selection and dimension reduction;
- Investigating how much additional predictive accuracy is gained once non-linearity is allowed.

The proposed models include OLS, Group LASSO, SCAD, Stepwise Selection with BIC criterion and Sliced Inverse Regression (SIR) among the linear methods and Random Forest and XGBoost among the non-linear ensemble ones.

Data Construction

The dataset has been constructed by statistically matching Bank of Italy's "*SHIW*" and ISTAT's "*Aspects of daily life*" to integrate both economic and non-economic determinants of well-being.

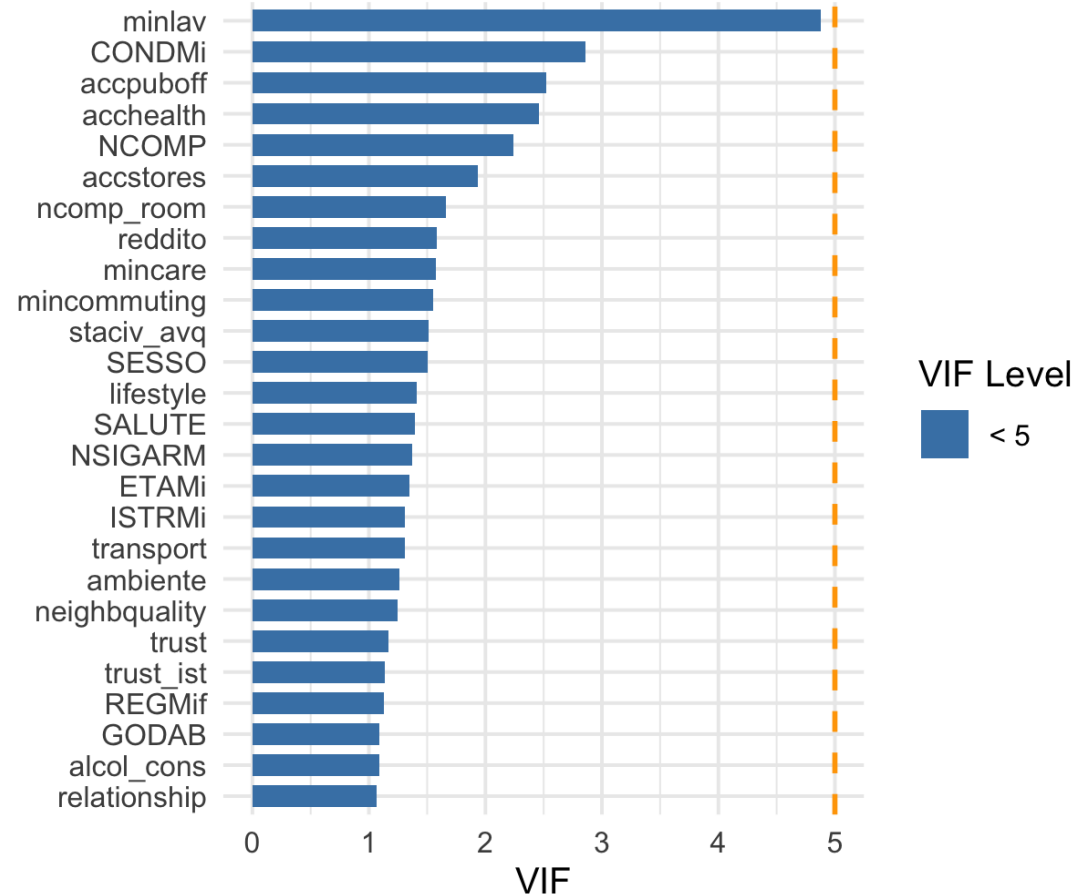
Indicators of wellbeing covering four different life domains (health and lifestyle, social relations, time allocation and housing conditions) have been created in "*Aspects of daily life*", the donor dataset, and transferred to SHIW using **nearest-neighbour hot-deck** with Gower distance computed on 8 socio-demographic features as similarity metric.

The matched dataset consists of 16.982 observations nested into 8.145 households, with 26 predictors.

The object of our prediction is the variable "VOTOVI", expressing life satisfaction of individuals on a 1-10 scale, which is assumed to be approximately continuous and normally distributed.

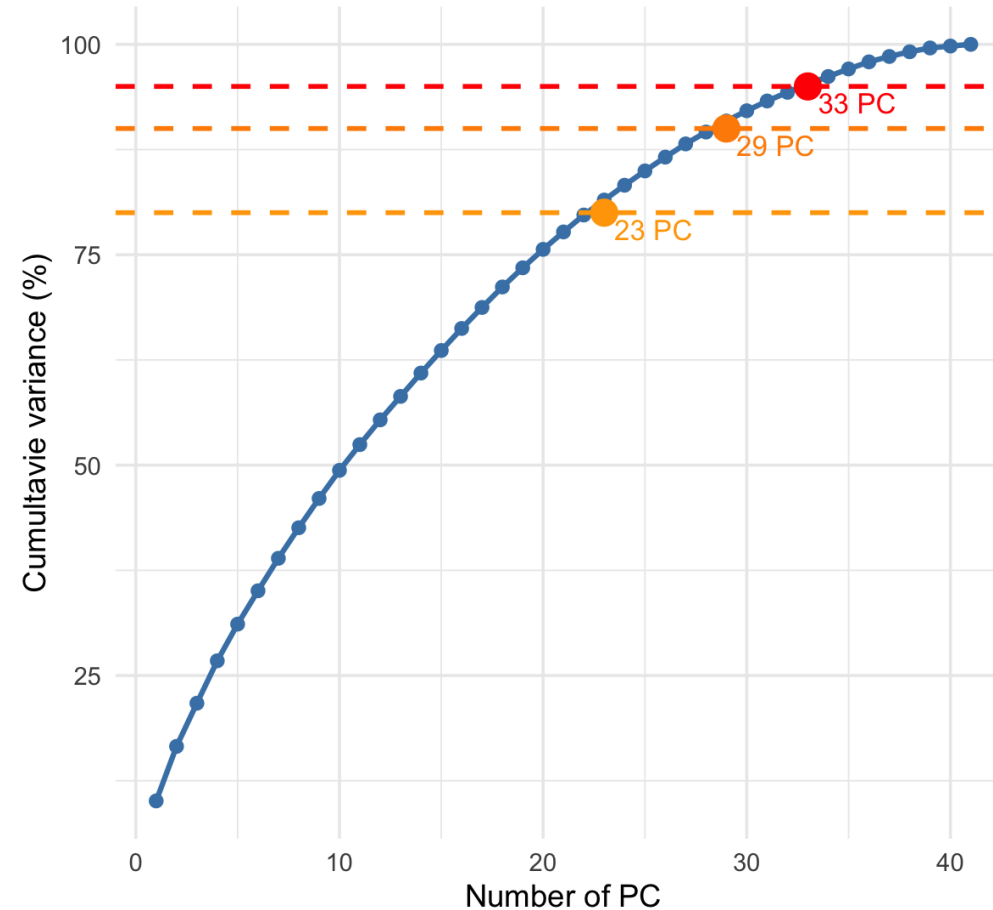
Preliminary Analysis

A preliminary stage of the analysis is the assessment of the correlation structure of the predictors, which is performed using **GVIF**, PCA and the correlation heatmap.



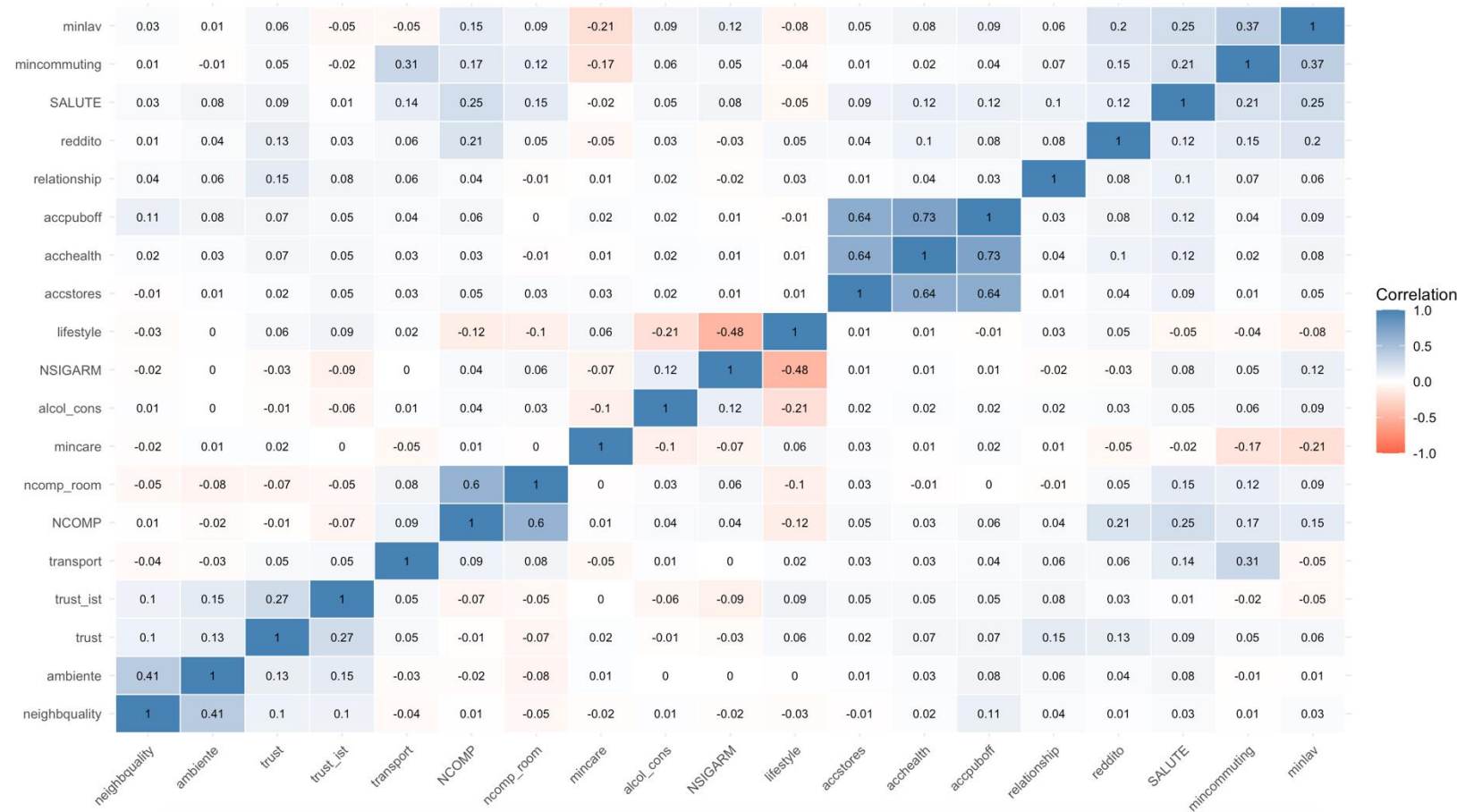
Preliminary Analysis

A preliminary stage of the analysis is the assessment of the correlation structure of the predictors, which is performed using GVIF, **PCA** and the correlation heatmap.



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Methods

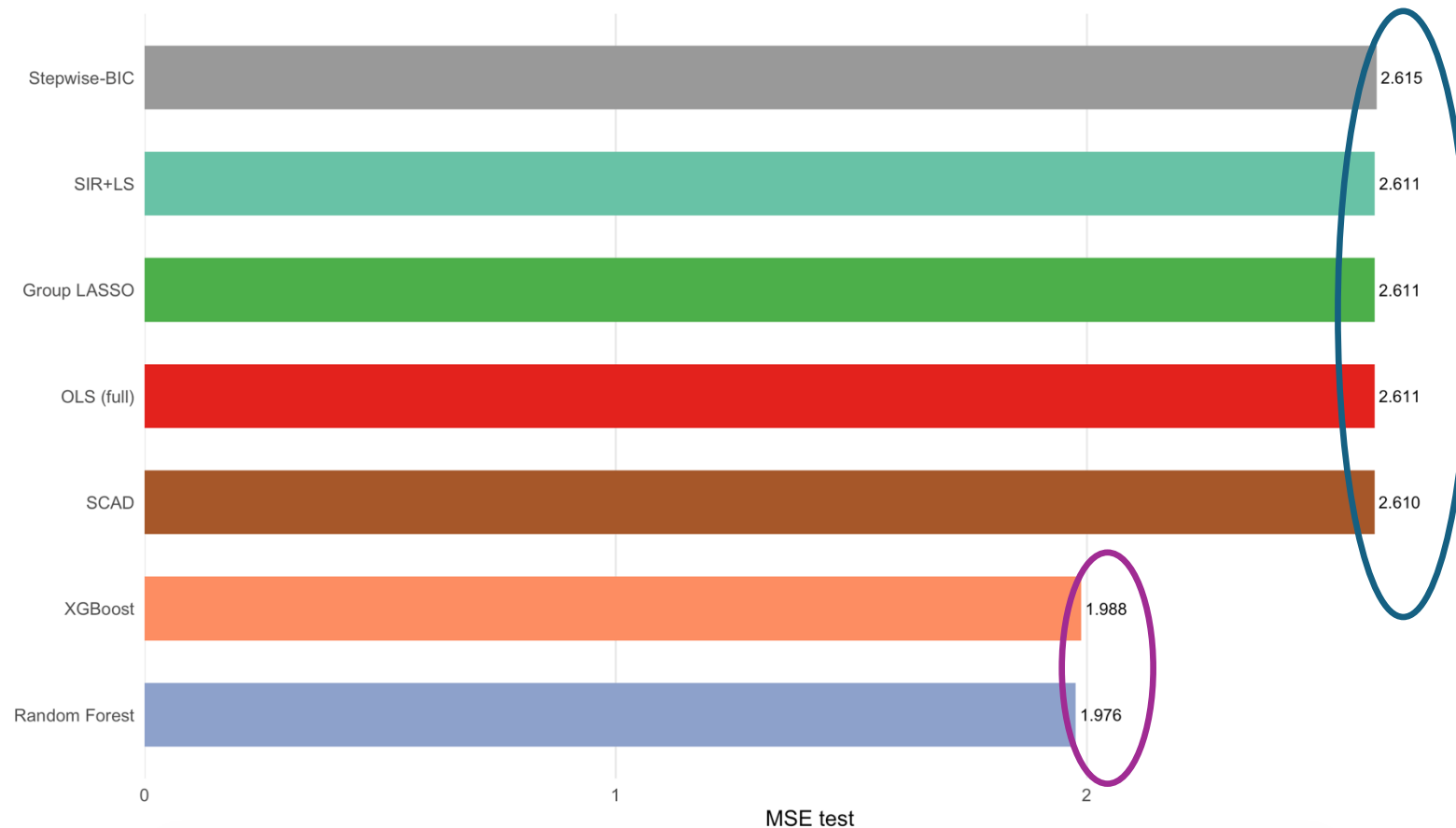
The main idea was to start from the most interpretable linear benchmark, then test whether the linear signal could be simplified through selection, shrinkage, or supervised dimension reduction, and finally compare these models with more flexible non-linear methods.



Since the data have a hierarchical structure (household → individual), train/test splitting and cross-validation folds were defined at the household level in order to avoid data leakage.

Results (1)

This figure highlights two core findings. **On the one hand**, the linear models all reach essentially the same predictive upper bound; **on the other hand**, more flexible non-linear methods achieve a clear jump in performance.



Results (2)

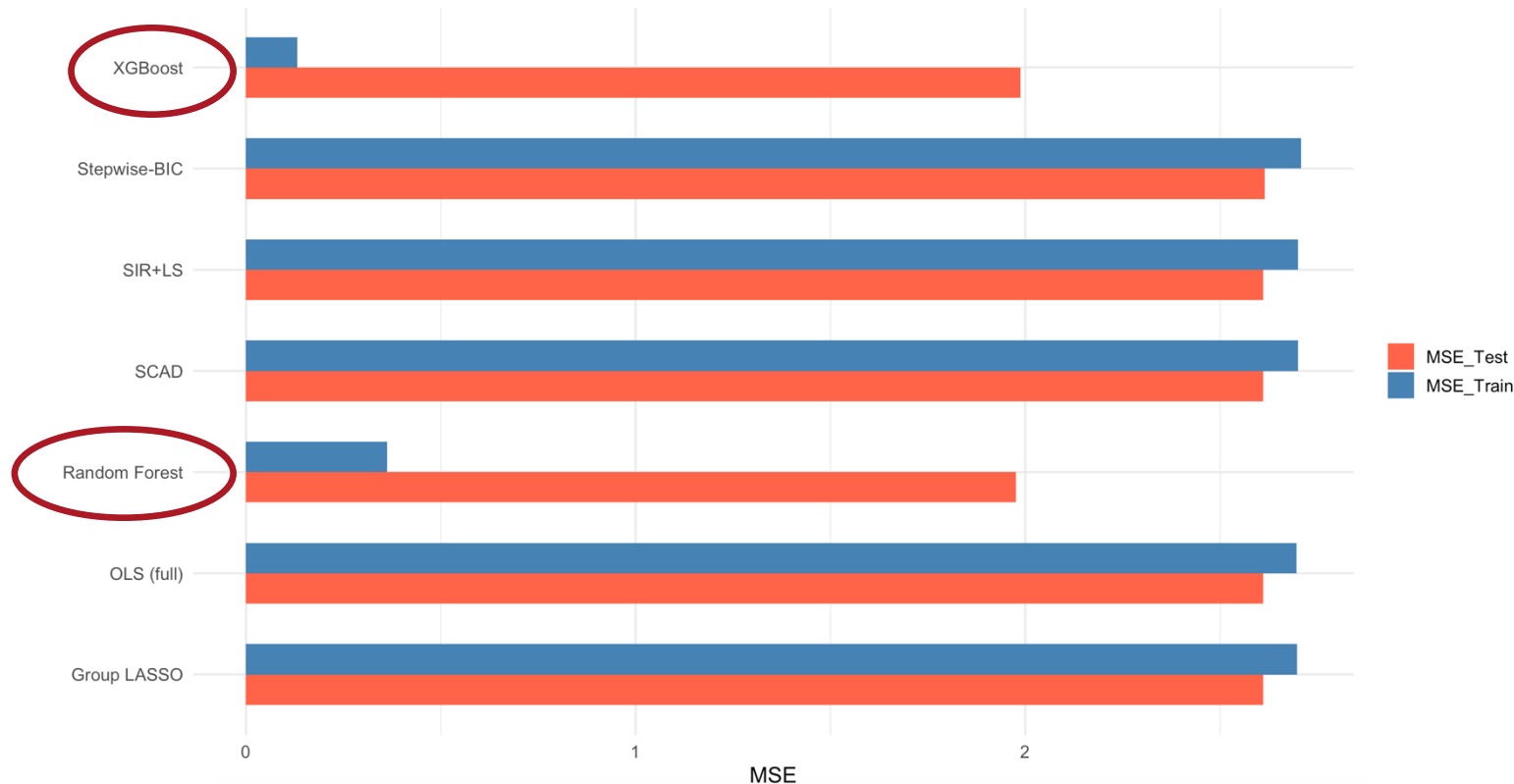
A second key result is that the **linear signal** can be **compressed** without any meaningful loss in predictive accuracy. Moving from the full OLS specification to sparser or lower-dimensional linear models leaves test-set MSE essentially unchanged, even though the representation becomes much more parsimonious.

This indicates that the relevant linear information is not spread uniformly across all predictors, but can be captured in a much smaller effective representation.

Model	MSE _{train}	MSE _{test}	Variables
OLS (full)	2.697	2.610	26
Group LASSO	2.697	2.611	24
SCAD	2.700	2.610	20
Stepwise-BIC	2.708	2.615	15
SIR + LS	2.699	2.611	9

Results (3)

The gain from flexible non-linear methods comes at the cost of a much larger train–test gap. Random Forest and XGBoost achieve better test performance, but they also fit the training sample much more aggressively than the linear models. This highlights the main trade-off of flexibility: better predictive power, but a **higher risk of overfitting**.



Additional Materials



Three other approaches were tested, without improvement in the predictive performance:

(1) Ridge Regression

(2) Dantzig Selector

(3) Expanded OLS with quadratic terms and pairwise interactions on top-10 RF variables

We conducted **robustness checks**:

(1) Changing the random seed

(2) Varying the train/test split ratio (60-40 and 80-20)

(3) Full ordinal model (dependent variable is categorical)

No additional method or robustness check produced evidence contradicting the main results reported before.

**Thank you
for attention!**



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